

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 2

What Numbers Actually Are

Exercises

Self-Study Curriculum

1. The quaternions extend the complex numbers. They have three imaginary units (i, j, k) instead of one, and multiplication is not commutative ($ij \neq ji$). Look up what operation quaternions represent geometrically (rotations in 3D). Why does losing commutativity make geometric sense when you move from 2D rotations to 3D rotations? (Hint: rotate a book 90° around the x-axis then 90° around the z-axis. Then do it in the other order.)

2. The rational numbers $\mathbb{Q}$ are “dense”, between any two rationals there is another rational, yet they are full of holes. Describe in your own words what it means for a set to be dense but not complete. Give a physical analogy (not the standard textbook one about a line with holes).

3. Consider the set $\{0, 1\}$ with addition defined modulo 2 (so $1+1=0$). This is a perfectly valid number system. What structural properties of the integers does it preserve? What does it lose? What kind of problem would this number system be useful for?

4. Run the diagonal argument yourself on a *specific* target: infinite binary strings (endless sequences

of 0s and 1s). A colleague claims to have a complete numbered list of all of them. Write down the recipe for the string that defeats the list, and state exactly which entry it differs from and where. Then explain why the same recipe *fails* if the colleague instead lists all *finite* binary strings: it must fail, since those genuinely are countable. Locating the exact step that breaks is the point of the exercise.
